Research CLI — Agent Request Architecture

packages/cli → packages/core routing → tools / MCP / research modules → model providers

Primary flow
Fallback / feedback
Dashed = architecture zone
Box = module + responsibility

PROJECT OVERVIEW

Research CLI is a terminal-first AI agent workflow that connects interactive sessions to models, tools, and MCP services.

Its core router manages context and provider selection while research modules cover writing, bibliography, analysis, and submission.

CLI FRONTEND · packages/cli

- **research.tsx** — interactive TUI
- **commands/** — mcp.ts; skills.tsx · hooks.tsx
- **ui/commands/research** — arxivCommand; paper · investigate · submit
- **acpSession.ts** — ACP session; stdio transport

CORE ORCHESTRATION · packages/core

Session

- **researchChat.ts** — researchRequest.ts
- **research-config-manager.ts** — academic settings
- **config / memory** — context window

Routing

- **modelRouterService** — routing/modelRouterService.ts
- **defaultStrategy** — routingStrategy.ts
- **classifierStrategy** — gemma · numerical
- **compositeStrategy** — approval · fallback override

TOOLS & RESEARCH MODULES

Workspace

- **edit.ts · grep.ts** — filesystem changes
- **shell / ls** — local execution

MCP

- **mcp-client.ts** — server sessions
- **mcp-client-manager.ts** — resource routing

analysis/

- **bibliography-manager** — citations
- **experiment-code-generator** — Python · R · Julia
- **research-data-analyzer** — statistics · ML

bibliography/

- **arxiv-mcp-client** — arXiv search
- **google-scholar-client** — scholar search
- **literature-investigator** — keyword-generator; reference-classifier

submission/

- **latex-manager** — project compile
- **journal-matcher** — venue selection
- **arxiv-latex-extractor** — template extract

loop

MODEL PROVIDERS

- **Gemini / Vertex**
- **OpenAI**
- **Anthropic**
- **Baidu ERNIE**
- **DeepSeek**
- **Qwen**
- **Groq / Mistral** — Cohere · Moonshot

EMBEDDED RUNTIMES

- **packages/sdk** — library embedding
- **packages/a2a-server** — agent executor; HTTP task API
- **vscode-ide-companion** — IDE companion
- **packages/devtools** — development tools

Primary path: prompt → research.tsx → modelRouterService → provider, with tools and research modules supplying observations.